Alvaro Gomariz

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ML Scientist with over 10 years of experience in machine learning for biomedical imaging and healthcare, including a PhD in Computer Vision and 5+ years in the pharmaceutical industry. I focus on identifying and realizing high-impact opportunities for ML in healthcare, from research to clinical trials, and on translating cutting-edge methods into robust, rigorously evaluated solutions. I have led cross-disciplinary projects that turn complex imaging and multimodal data into evidence for decision-making.

Experience

Roche, Switzerland **Mar 2021 - Present**

Senior Data scientist *Oct 2023 - Present*

- Lead the technical and scientific direction of the AI video colonoscopy program, guiding a core team of 3-5 AI/ML scientists and 2-4 contract engineers across gap analyses, rigorous model evaluation, MLOps architecture, and prioritized algorithm roadmaps
- Partner with biostatistics and clinical science to build a multimodal prognostic model (endoscopy imaging + clinical data) that outperforms the clinical baseline, evaluated for covariate adjustment in the statistical analysis plan of AMETRINE-1 and 2 (Phase 3 IBD trial)
- Lead an annotation campaign with 5 academic experts and 4 gastroenterologists to benchmark endoscopic scoring reliability and develop AI for disease-severity scoring and lesion/feature detection that matches inter-rater agreement; implement ETL pipelines and automated QC of annotations
- Curate and harmonize multimodal patient data (real-world and clinical-trial), and build an MCP interface with LLM agents that lets the team query our relational Athena database without SQL expertise
- Publish results, file a patent application, and present at conferences, establishing partnerships with Scribe (medical annotation, selected for cost and operational efficiency) and Virgo (image-analysis evaluation and a potential multi-sponsor consortium)
- Until March 2024, led a matrix team of 5-10 AI/ML scientists building deep learning capabilities in digital pathology, contributing to a company-wide foundation model initiative and producing an internal software package and three papers as lead author
- Co-lead a data science network of 25-35 scientists, fostering a connected community for knowledge sharing and capability building
- Side project: Build an LLM agent (ScrapeGraphAI, LangChain, RAG) compiling Roche suppliers' ISO and sustainability credentials from public sources with source links, replacing an hour of manual search per supplier

Data scientist *Aug 2022 - Sep 2023*

- Built a proof-of-concept for lung cancer patient stratification from H&E slides, targeting signal comparable to routine genetic screening
- Addressed severe class imbalance and sparse signal in gigapixel images through data stratification, benchmarking, and evaluation of ML methodologies
- Steered and supervised the development of novel ML solutions (multiple instance learning, graph neural networks, self-supervised learning / foundation models, survival analysis) and software development (MLops)
- Designed rigorous benchmarking frameworks to evaluate internal models against external vendors, directly guiding corporate build-vs-buy decisions
- Strengthened the group's technical foundation by identifying reusable ML methodologies across products and organizing matrix teams around them

Postdoctoral researcher *Mar 2021 - Jul 2022*

- Developed and published methods (semi-supervised learning, contrastive CNNs, unsupervised domain adaptation) improving ophthalmic OCT segmentation across devices, enabling predictions on new devices without additional annotations
- Prototyped and informed novel methodologies for internal OCT analysis software (momaku, ROSA), extending it to new devices and real-world data, and informing subsequent foundation model efforts

Technical lead - MicroscopeIT, Poland & U.S.A. (remote) **Mar 2018 - Mar 2019**

- Led R&D project from conception to delivery, creating an image analysis product leveraging deep learning and spatial statistics for large 3D microscopy clinical datasets
- Directed a team of 3-5 engineers to develop the software for biotechnology clients, including clinicians and researchers

Doctoral researcher - ETH & University Hospital Zurich, Switzerland Sept 2016 - Feb 2021

- Conducted research on deep learning methods (attention networks, learning with missing modalities, uncertainty estimation) to facilitate image analysis tasks (segmentation, detection, regression, and classification) of large 3D microscopy datasets, leading to multiple publications in renowned scientific journals
- Collaborated with biologists, clinicians, and healthcare companies, on multiple imaging domains (MRI, ultrasound, micro-CT, and different microscopy types), including software development for users with no programming experience, leading to scientific publications and research funding
- Wrote research grants resulting in project funding, supervision of 7 students, teaching, and organization of seminars

Education

Doctor of Science in Computer Vision - ETH Zurich, Switzerland Oct 2016 - Feb 2021

- Best PhD Award from the Center for Experimental and Clinical Imaging Technologies (EXCITE) Zurich for outstanding thesis in the field of biomedical imaging
- PhD thesis: "Quantitative Fluorescence Microscopy with Marker-Efficient and Uncertainty-Aware Segmentation and Detection Using Deep Learning"

MSc in Biomedical Engineering - ETH Zurich, Switzerland Oct 2014 - Sep 2016

- 1st of class: awarded Willi-Studer Prize
- Master thesis: "Applying 3D quantitative microscopy to study global topography and cellular interactions in the bone marrow"
- Semester project: "Investigation of image registration for sliding boundaries"

BSc in Biomedical Engineering - University Carlos III of Madrid, Spain Oct 2010 - Sep 2014

- 1st of class: Awarded Extraordinary End of Studies Award
- Bachelor thesis: "Resolution study of optical tomography systems in microscopy"

Skills

Machine learning	Foundation models, multi-modality, distributed pre-training, GNNs, uncertainty estimation
Software development	MLOps, data engineering, CI/CD, ETL, high performance computing (SLURM & LSF), git
Programming & Scripting	Python, SQL, Matlab, bash/shell, Docker, R
Frameworks & tools	PyTorch, TensorFlow, AWS, V7, Nextflow, SciPy Ecosystem
Imaging tools	Fiji, Napari, 3D Slicer, ITK, OpenCV
Generative & Agentic AI	LLM APIs, ScrapeGraphAI, LangChain, MCP, AI-assisted development (Claude Code, Cursor, Copilot)
Project leadership	Matrix teams, strategic planning, product development, cross-functional collaboration
Agile certifications	ICAgile Certified Professional (May 2024), Professional Scrum Master^{TM} I (Aug 2022)
Languages	English (fluent), Spanish (native), German (basic)

Additional information

Scientific Contributions

Journal articles

- **Gomariz A**, Kikuchi Y, Li Y, Albrecht A, Maunz A, Ferrara D, Lu H, Goksel O, "Joint semi-supervised and contrastive learning enables domain generalization and multi-domain segmentation", *Medical Image Analysis* 103, p.103575 (2025). doi: 10.1016/j.media.2025.103575
- Yamada Y, Nguyen TT, Impellizzieri D, Mineura D, Shibuya R, **Gomariz A**, Haberecker M, Nilsson J, Nombela-Arrieta C, Jungraithmayr W, Boyman O "Biased IL-2 signals induce Foxp3-rich pulmonary lymphoid structures and facilitate long-term lung allograft acceptance in mice", *Nature Communications* 1383 (14) (2023). doi: 10.1038/s41467-023-36924-z
- Vijaykumar A, Helbling PM, Thelen F, Fazio S, Mun Y, Pereira AL, Zielińska KA, Buschl P, Zerkatje T, Loosli K, Isringhausen S, Menze B, Nagasawa T, Roeder I, **Gomariz A**, Manz MG, Yokomizo T, Nombela-Arrieta C "Tissue-scale mapping reveals a central role of hepatoblasts in the regulation of fetal liver hematopoiesis and stem cell maintenance", *Developmental Cell* 61, 901-918.e4 (2026). doi: 10.1016/j.devcel.2026.01.011
- **Gomariz A**, Portenier T, Nombela-Arrieta C, Goksel O, "Probabilistic Spatial Analysis in Quantitative Microscopy with Uncertainty-Aware Cell Detection using Deep Bayesian Regression", *Science Advances* 8 (5) (2022). doi: 10.1126/sciadv.abi8295
- **Gomariz A**, Portenier T, Helbling PM, Isringhausen S, Suessbier U, Nombela-Arrieta C, Goksel O, "Modality Attention and Sampling Enables Deep Learning with Heterogeneous Marker Combinations in Fluorescence Microscopy", *Nature Machine Intelligence* p. 1-13 (2021). doi: 10.1038/s42256-021-00379-y
- Isringhausen S, Mun Y, Kovtonyuk L, Krautler N, Suessbier U, **Gomariz A**, Spaltro G, Helbling PM, Wong HC, Nagasawa T, Manz MG, Oxenius A, Nombela-Arrieta C, "Chronic viral infections persistently alter marrow stroma and impair hematopoietic stem cell fitness ", *Journal of Experimental Medicine* 218 (12) (2021). doi: 10.1084/jem.20192070
- Ghazi S, Bourgeois S, **Gomariz A**, Bugarski M, Haenni D, Martins JR, Nombela-Arrieta C, Wagner CA, Hall AM, Eilidh C, "Multiparametric imaging reveals that mitochondria rich intercalated cells in the kidney collecting duct have a very high glycolytic capacity", *The FASEB journal* 34 (6), pp. 8510–8525 (2020). doi: 10.1096/fj.202000273R
- **Gomariz A**, Isringhausen S, Helbling PM, and Nombela-Arrieta C, "Imaging and spatial analysis of hematopoietic stem cell niches", *Annals of the New York Academy of Sciences* 1466(1):5-16 (2019). doi: 10.1111/nyas.14184
- Hess M, **Gomariz A**, Goksel O, Ewald CY, "In-vivo quantitative image analysis of age-related morphological changes of C. elegans neurons reveals a correlation between neurite bending and novel neurite outgrowths", *eNeuro* 6(4) (2019). doi: 10.1523/ENEURO.0014-19.2019
- Agarwal P, Isringhausen S, Li H, Paterson AJ, He J, **Gomariz A**, Nagasawa T, Nombela-Arrieta C, and Bhatia R, "Mesenchymal niche-specific expression of Cxcl12 controls quiescence of treatment-resistant leukemia stem cells", *Cell Stem Cell* 24 (5), 769–784.e6 (2019). doi: 10.1016/j.stem.2019.02.018
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Becker A, Boss A, Nagasawa T, Paul G, Goksel O, Székely G, Stoma S, Nørrelykke SF, Manz MG, and Nombela-Arrieta C, "Quantitative spatial analysis of haematopoiesis- regulating stromal cells in the bone marrow microenvironment by 3D microscopy", *Nature Communications* 9 (1) (2018). doi: 10.1038/s41467-018-04770-z
- Takizawa H, Fritsch K, Kovtonyuk LV, Saito Y, Yakkala C, Jacobs K, Ahuja AK, Lopes M, Hausmann A, Hardt WD, **Gomariz A**, Nombela-Arrieta C, and Manz MG, "Pathogen-Induced TLR4-TRIF Innate Immune Signaling in Hematopoietic Stem Cells Promotes Proliferation but Reduces Competitive Fitness", *Cell Stem Cell* 21 (2), 225–240.e5 (2017). doi: 10.1016/j.stem.2017.06.013

Conference papers

- Blasco Fernandez P, Bossuyt P, Daperno M, Kopylov U, Bouhnik Y, Karmiris K, Benmansour F, Gutierrez Becker B, Fraessle S, Stimpel B, Levitte S, **Gomariz A** "Quantifying Inter-Rater Reliability of Mucosal Features in Ulcerative Colitis Endoscopy", *Journal of Crohn's and Colitis* Volume 20, Issue Supplement_1 (2026). doi: 10.1093/ecco-jcc/jjaf231.052
- **Gomariz A**, Gutierrez-Becker B, Girycki R, Szostak P, Stimpel B, Levitte S, Abbasi-Sureshjani S "Assessing the Impact of Biopsy Artifacts on AI-Based Endoscopic Scoring for Ulcerative Colitis", *Journal of Crohn's and Colitis* Volume 20, Issue Supplement_1 (2026). doi: 10.1093/ecco-jcc/jjaf231.699
- Albuquerque T, Yuce A, Herrmann MD, and **Gomariz A**, "Characterizing the Interpretability of Attention Maps in Digital Pathology", *arXiv* (2024). doi: 10.48550/arXiv.2407.02484
- Bredell G, Fischer M, Szostak P, Abbasi-Sureshjani S, and **Gomariz A**, "The Importance of Downstream Networks in Digital Pathology Foundation Models", *International Workshop on Foundation Models for General Medical AI - MICCAI* pp. 10-19 (2024). doi: 10.1007/978-3-031-73471-7_2
- Ibañez V, Szostak P, Wong Q, Korski K, Abbasi-Sureshjani S, and **Gomariz A**, "Integrating multiscale topology in digital pathology with pyramidal graph convolutional networks", *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 1-4 (2024). doi: 10.1109/ISBI56570.2024.10635702
- Kikuchi Y, **Gomariz A**, Li Y, Lu H, Albrecht A, Ferrara D, "Device adaptation of optical coherence tomography (OCT) retinal layer segmentation algorithm using unlabeled target data", *Investigative Ophthalmology & Visual Science, ARVO abstract (2023)*
- **Gomariz A**, Lu H, Li Y, Albrecht A, Maunz A, Benmansour F, Valcarcel AM, Luu J, Ferrara D, Goksel O, "Unsupervised Domain Adaptation with Contrastive Learning for OCT Segmentation", *International Conference on Medical image computing and computer-assisted intervention (MICCAI)* pp. 351-361 (2022). doi: 10.1007/978-3-031-16452-1_34
- Li Y, Lu H, Albrecht T, Maunz A, Neubert A, Benmansour F, **Gomariz A**, Luu J, Ferrara D "Deep learning (DL) to segment retinal layer disruption on optical coherence tomography (OCT) in neovascular age-related macular degeneration (nAMD)", *Investigative Ophthalmology & Visual Science, ARVO abstract (2022)*
- **Gomariz A**, Lu H, Li Y, Albrecht A, Maunz A, Benmansour F, Luu J, Goksel O, Ferrara D, "A unified deep learning approach for OCT segmentation from different devices and retinal diseases", *Investigative Ophthalmology & Visual Science, ARVO abstract (2022)*
- **Gomariz A**, Egli R, Portenier T, Nombela-Arrieta C, and Goksel O, "Utilizing Uncertainty Estimation in Deep Learning Segmentation of Fluorescence Microscopy Images with Missing Markers", *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 371-374 (2021). doi: 10.1109/ISBI48211.2021.9434158
- **Gomariz A**, Li W, Ozkan E, Tanner C, and Goksel O, "Siamese Networks with Location Prior for Landmark Tracking in Liver Ultrasound Sequences", *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 1757-1760 (2019). doi: 10.1109/ISBI.2019.8759382
- Goksel O, Vishnevsky V, **Gomariz A**, and Tanner C, "Imaging of Sliding Visceral Interfaces During Breathing", *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 298-301 (2016). doi: 10.1109/ISBI.2016.7493268